

RooMatch

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Motivation and Novelty

Online real estate and short-term rental platforms often contain duplicate or misleading listings of the same property presented with different furniture, decor, or staging. These appearance changes make it difficult for vision models to recognize that images depict the same physical room.

Since no public dataset captures the same room under controlled interior variations, this project introduces a novel synthetic dataset and retrieval framework that focuses on matching rooms based on invariant spatial and structural cues rather than superficial visual changes.

Real Image:



Dataset Example

Generated
Synthetic:



Dataset Room Generation Prompt Structure

```
"furniture": [  
    "replace the sofa with a different style sofa",  
    "add a small coffee table in front of the sofa",  
    "remove one piece of furniture from the room"  
],  
"decor": [  
    "add decorative pillows and throws",  
    "add framed artwork on the walls without changing wall positions",  
    "add small decorative objects on shelves and tables"  
],  
"plants": [  
    "add one medium indoor plant in a corner",  
    "add several small potted plants on surfaces",  
    "remove all indoor plants"  
],  
"people": [  
    "add one person sitting casually on the sofa",  
    "add a person standing naturally in the room"  
],  
"clutter": [  
    "make the room very tidy and minimal",  
    "add light everyday clutter like books and magazines",  
    "slightly rearrange small objects on surfaces"  
],  
"materials": [  
    "change fabric materials of furniture only",  
    "change color of pillows and textiles",  
    "change furniture upholstery while keeping shapes identical"
```

```
BASE_PROMPT = (  
    "Photorealistic INPAINTING of the SAME physical room as the input image. "  
    "This is an interior edit, NOT a new image. "  
    "Keep EXACT room structure: same walls, same wall positions, "  
    "same windows, same doors, same ceiling, same floor, same camera position. "  
    "Do NOT add, remove, resize, move, or invent windows, doors, openings, or walls. "  
    "Only modify movable interior objects such as furniture, decor, people, plants, "  
    "and small items. "  
    "The room layout and geometry must remain IDENTICAL to the input image."  
)
```

```
NEGATIVE_PROMPT = (  
    "new room, different room, room redesign, architectural changes, "  
    "changed layout, changed geometry, perspective change, camera move, "  
    "added window, removed window, extra window, fake window, "  
    "added door, removed door, extra door, fake door, "  
    "wall opening, doorway, archway, hallway, "  
    "structural changes, walls moved, walls removed, "  
    "ceiling changed, floor changed, "  
    "distorted walls, warped geometry, unrealistic perspective, "  
    "low quality, blurry, artifacts"
```

Room Matching Dataset

Dataset Description

- The dataset contains **1000** real room images collected from various sources.
- For each real room image **10** synthetic variants were generated with **Stable Diffusion Inpainting** to increase diversity while preserving the same room layout.
- Each image is assigned a **room ID**, which acts as the ground truth for evaluation.

Dataset Preprocessing

- All images are converted to **RGB** and resized to a fixed resolution of **512×512 pixels** to ensure consistency across the dataset.
- **Random masks** focus on central and lower regions of the image, where furniture and decor typically appear.
- Attribute modifications are **randomly** sampled.

Data Labeling

- Room attributes include materials, people, furniture, decor, plants, and clutter.
For each room attribute there are 3 possible outcomes.
(we tried curtains, wall decor but they affected the room structure)
- **BASE_PROMPT** ensures the same room geometry is preserved while decor, materials, and movable objects are varied.
- **NEGATIVE_PROMPT** prevents generating entirely different rooms or altering the room layout.

Room Matching Model

We use **DINOv2 (ViT-B/14)**, a self supervised vision transformer pretrained on large scale natural images, as the feature extractor for room embeddings. Images are embedded into a normalized latent space and compared using cosine similarity for image retrieval.

To adapt the model to room level matching, we fine-tune only the last transformer block using **Triplet Loss**. Each training sample consists of an anchor image, a positive image from the same room, and a negative image from a different room. This encourages embeddings of the same physical room to cluster tightly while pushing different rooms further apart.

Results Examples:

Recall@0 (failure):

Query index: 407, Room ID: 48

Query (Room 48)	Room 45	Room 46	Room 45	Room 45	Room 4	Room 48	Room 48	Room 4	Room 4	Room 42
										

Recall@3:

Query index: 342, Room ID: 42

Query (Room 42)	Room 42	Room 42	Room 42	Room 46	Room 45	Room 45	Room 42	Room 4	Room 42	Room 42
										

Recall@7:

Query index: 88, Room ID: 2

Query (Room 2)	Room 2	Room 2	Room 2	Room 2	Room 2	Room 2	Room 2	Room 32	Room 0	Room 2
										

Recall@10:

Query index: 14, Room ID: 24

Query (Room 24)	Room 24	Room 24	Room 24	Room 24	Room 24	Room 24	Room 24	Room 24	Room 24	Room 24
										

Metrics & Accomplishments

Metrics:

- **Strict Recall@K** measures the percentage of queries for which **all top-K** retrieved images belong to the same room ID as the query image, enforcing a strict requirement that every retrieved neighbor **in a row** must be correct.
- **Intra-mean** is the average distance between different stylistic variations of the **same** physical room.
A lower value indicates better consistency and recognition of the core room identity despite changes.
- **Inter-mean** The average distance between **distinct** physical rooms.
A higher value signifies that the model successfully differentiates between unique architectural spaces.
- **gap** The critical **separation** margin between the Intra-mean and Inter-mean. (inter - intra)
A wider gap is the ultimate indicator of the model's success.

Achievements:

- **Demonstrated measurable improvement** in room-level image retrieval through fine-tuning.
- **Increased embedding separability**, achieving a significantly larger intra–inter distance gap.
- **Validated robustness** using a strict evaluation protocol (Strict Recall@K).

Methodology

Project process:

- Generated a controlled dataset: 1 real room image + 10 synthetic variants per room.
- Built a room image retrieval pipeline using DINOv2 embeddings.
- Fine-tuned the model using metric learning (Triplet Loss).
- Extracted and normalized embeddings for all images.

Data & Evaluation:

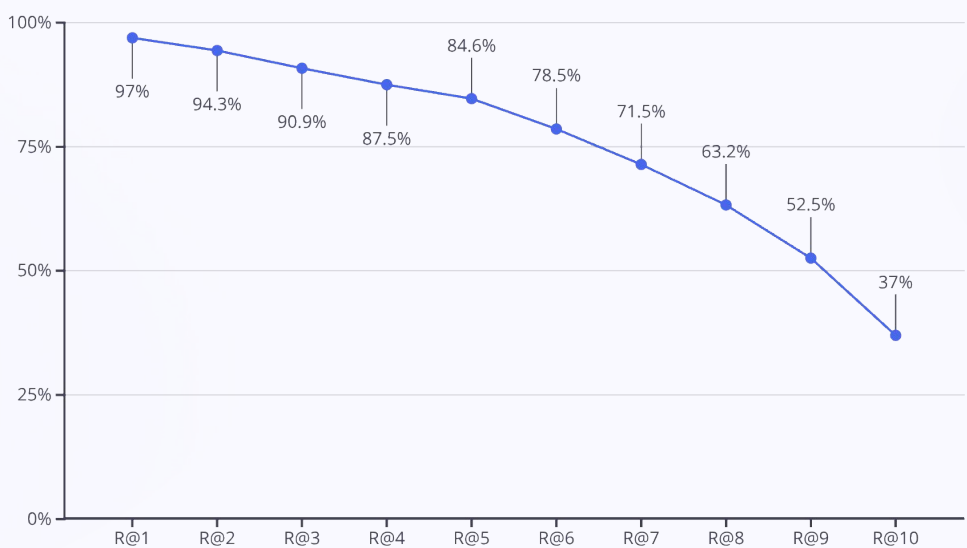
- **Data:** 1,000 real room images, 10,000 synthetic variants (same room, altered content).
- **Model:** DINOv2 ViT-B/14 (base vs. fine-tuned).
- **Metrics:**
 - ☐ Strict Recall@K
 - ☐ Intra-room vs. inter-room distance analysis
 - ☐ Embedding gap (inter – intra)

Efforts done:

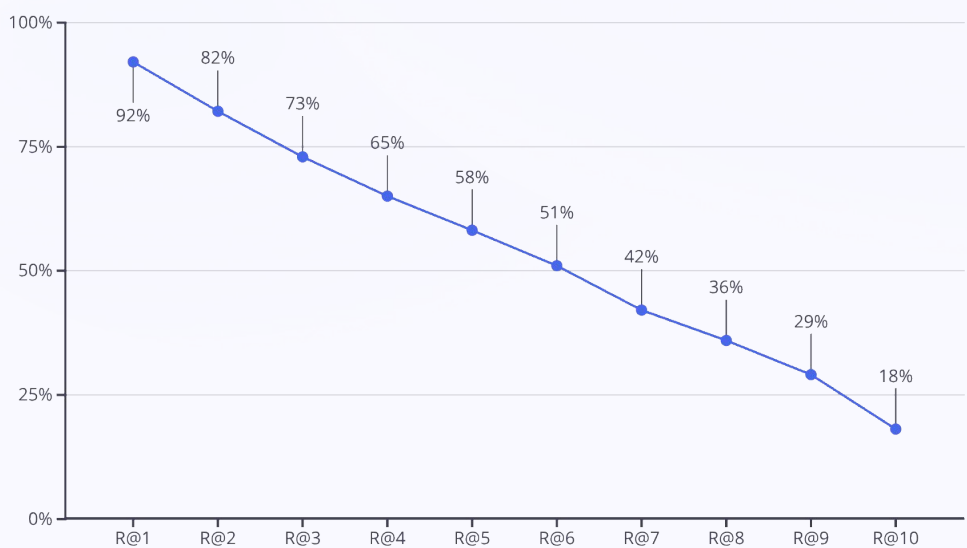
- Designed **strict evaluation** to avoid inflated performance.
- Used **controlled synthetic variations** to enforce invariance learning.

Results

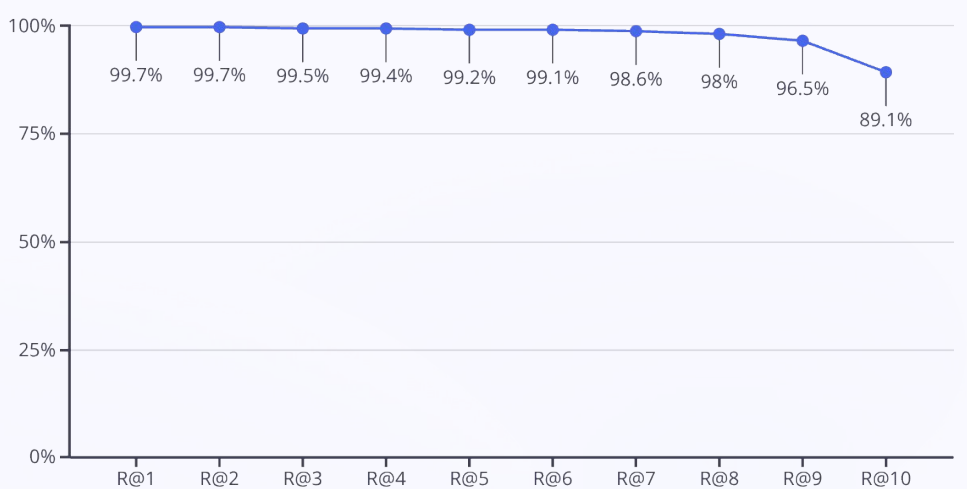
Base Model Results:



Interm Results:



Fine-tuned Model Results:



Embedding separability:

Model	intra_mean	inter_mean	gap
Base Dinov2	0.1923	0.7809	0.5885
Fine-tuned	0.0858	0.8367	0.7509

Conclusions

Achieved results:

- Fine-tuning significantly improved room level retrieval quality.
- Achieved tighter intra-room clustering and stronger separation between rooms.
- Improvements were consistent under Strict Recall@K, a highly challenging metric.

Things we learnt:

- Generic pretrained embeddings were insufficient for room level retrieval without adaptation.
- Some visually similar rooms remain challenging under strict evaluation.

Future Ideas:

- Analyze performance differences across room categories (bedroom, kitchen, living room, bathroom).
- Test synthetic-to-real retrieval as a robustness benchmark.